

Week 40 exercises – INF201 2026

Task 1 – Set up a GitHub repository

Follow these steps to create your (first?) GitHub repository:

1. If you don't have one already, create an account on GitHub
 - a. I recommend using your personal email address as primary email address and adding your NMBU email address as secondary email address, to ensure access also after you leave NMBU
2. Download and install git from <https://git-scm.com/install/>
 - a. Configure your identity in your VS Code integrated terminal (or your regular terminal):

```
git config --global user.name "Your Full Name"
```

```
git config --global user.email "your_github_email@example.com"
```

Note that the email address must match your primary GitHub email address exactly

3. Also in the terminal, ensure that new repositories default to main

```
git config --global init.defaultBranch main
```
4. Connect VS Code to your GitHub account
 - a. Open VS Code.
 - b. Click on the Accounts icon (bottom-left gear or avatar).
 - c. Select Sign in with GitHub and authorize VS Code in your browser.
5. Create and push your first repository
 - a. Create a new public repository on GitHub (in your web browser) named git-practice.
 - b. Check the box to Add a README.md file
 - c. Check the box to Add .gitignore and select the Python .gitignore template.
 - d. In VS Code, open the Command Palette (Ctrl+Shift+P / Cmd+Shift+P), type `Git: Clone`, and paste your repository URL.
 - e. Open the cloned folder in VS Code.
 - f. Create a file named `hello.py` with `print("Hello, Git!")`.
 - g. Use the VS Code Source Control tab (or terminal) to stage `hello.py`, enter the commit message "Add hello script", and push your changes to GitHub.
 - h. Note: Do NOT upload files by dragging and dropping them into the GitHub web interface.
6. Verify and submit your repository
 - a. Visit your repository URL on github.com in a browser.

- b. Verify that hello.py appears in the file list and displays your commit message.
- c. Submit your repository URL to Canvas, either directly as a link or a link pasted into a txt file
- d. Note that the repository must be public for us to be able to review it. You may change it to private after we have graded the assignment.

Optional, if you have time and want to get some more Git practice: Work through this tutorial: [Introduction to Git - Training | Microsoft Learn](#)

Task 2 – Working with multiple formats

You are building a quick script to check laboratory equipment calibration status. You will read target settings from a YAML file, look up sensor locations in an Excel sheet, process calibration logs from a CSV file, and output the out-of-date sensors to a JSON file.

Write a script check_sensors.py that executes the following pipeline:

- Read settings:
 - Read config.yml and get max_days_since_calibration and output_file
- Read and join data:
 - Read sensors.xlsx
 - Read calibrations.csv
 - Match each sensor's location/owner with its calibration status.
- Filter and export:
 - Identify sensors where days_since_calibration > max_days_since_calibration.
 - Export the overdue sensors as a formatted JSON array to the filename specified in config.yml using json.dump(..., indent=2).
 - This json file should include sensor_id, lab/owner information as well as days_since_calibration for the overdue sensors only
- Submit the script and the generated json file either as usual on Canvas, or in a github repository (must be public, submit the link to the repository)

Hand-in information

- Remember to include filename extension for all submitted files
- You may submit Python scripts (.py) or Jupyter notebooks (.ipynb)
- If you submit a Jupyter notebook, please upload a pdf version as well
- AI chat-log can be submitted in its entirety or summarized (see lecture_02)

